Statistical Analysis and Modeling of Inter-Satellite PRN Code Interference in GPS L1 Signals

Nancy Abraham

University of New Brunswick

Outline

Introduction

Motivation

Theoretical Background

Methodology

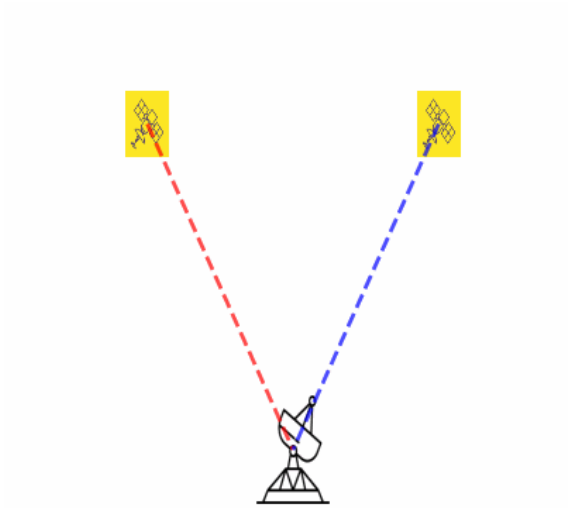
Results

Model

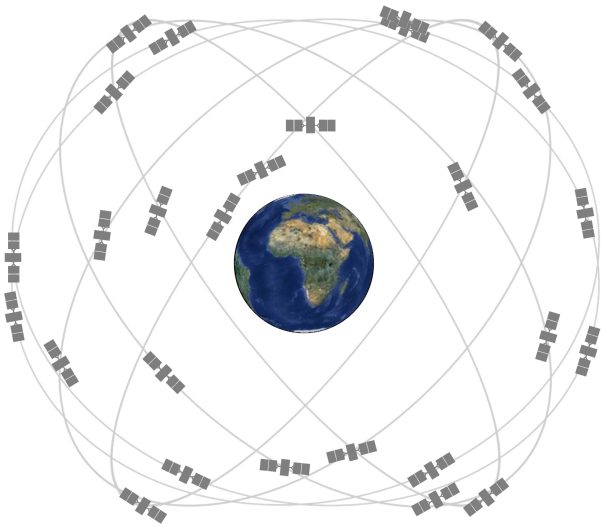
Conclusion

Introduction

- GPS provides global positioning and timing services.
- Signals suffer interference, including PRN code cross-correlation.
- This study identifies and models interference signatures to support improved in GNSS signal filtering.



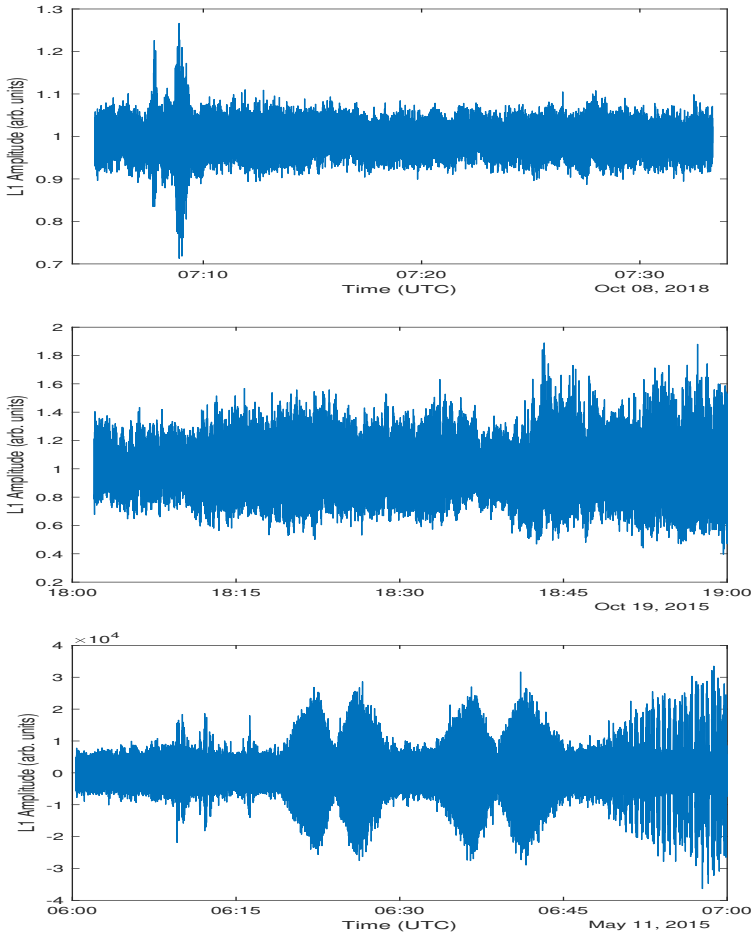
Animated illustration of PRN code interference from dual satellite paths.



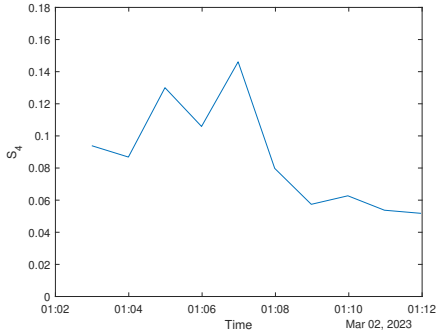
GPS constellation. Source: GPS.gov

Motivation

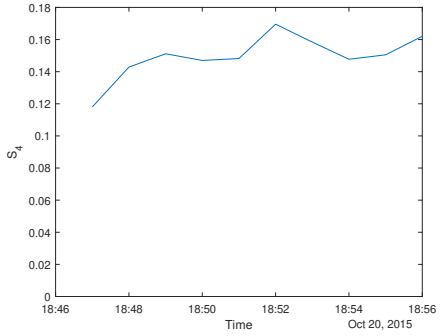
- GPS signal fluctuations are commonly attributed to ionospheric scintillation or multipath.
- However, inter-satellite interference due to PRN code cross-correlation remains underexplored.



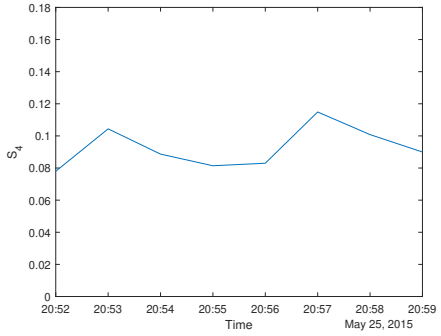
Amplitude fluctuations: (Top) Scintillation, (Middle) Multipath, (Bottom) Interference



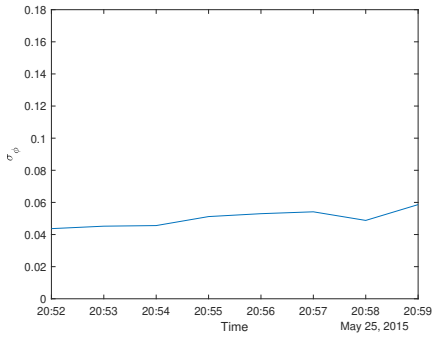
a. Scintillation Amplitude.



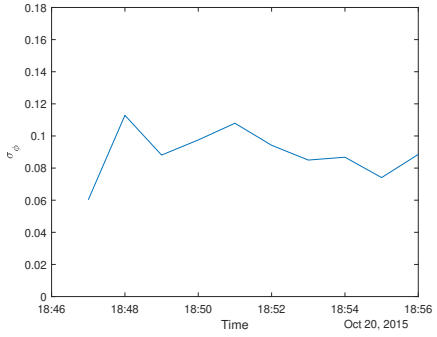
b. Multipath Amplitude.



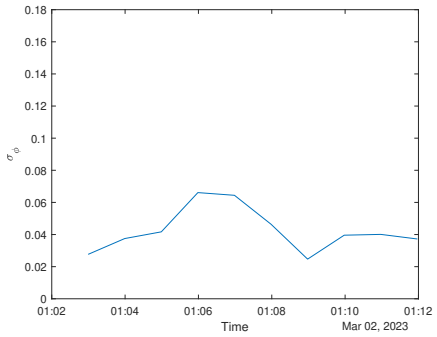
c. Interference Amplitude.



d. Scintillation Phase.



e. Multipath Phase.



f. Interference Phase.

- S_4 (intensity) and σ_ϕ (phase) are standard scintillation indices.
- Similar values can occur during multipath and inter-satellite interference.
- Thus, relying solely on these indices can lead to misclassifications.

Research Objective

- **Identify** GPS L1 C/A code inter-satellite interference and differentiate it from scintillation and multipath.
- **Statistically characterize** interference events based on observed Doppler shifts, PRNs involved, event durations, and frequency overlap patterns.
- **Apply and validate** a frequency-based model to analyze interference behavior using observational data and simulation.

Theoretical Background

- GPS L1 signals use C/A codes, which are short repeating Gold codes with a 1 ms period.
- These codes are designed to let multiple satellites share the same frequency using CDMA.
- Each PRN code has strong autocorrelation properties, helpful for tracking and acquisition.
- However, the cross-correlation between different PRNs is low but not zero.
- This low-level interference is usually negligible, but can become significant in specific conditions.

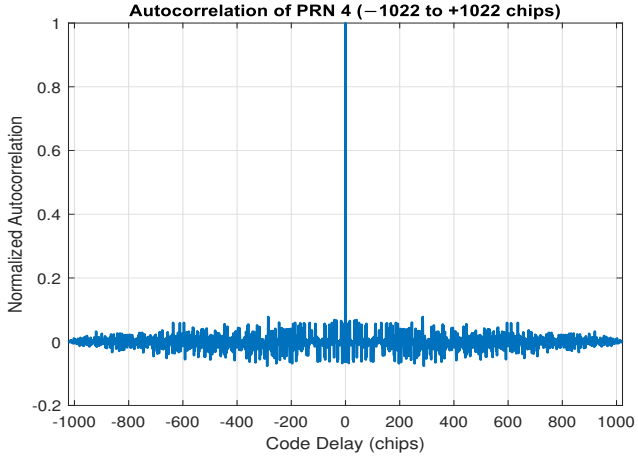


Fig: Autocorrelation of PRN 4.

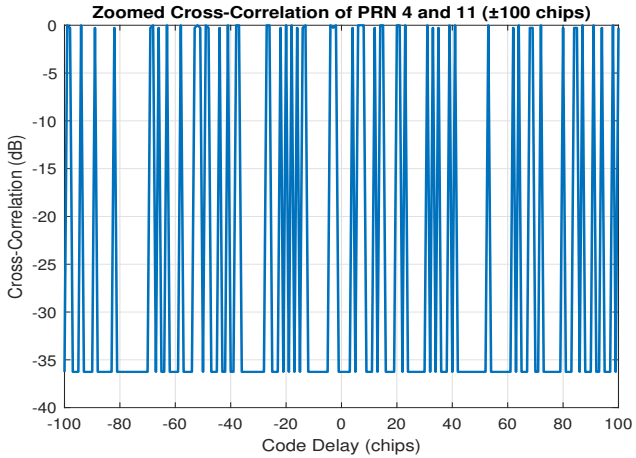


Fig: Cross-correlation between PRN 4 and PRN 11.

- GPS C/A codes exhibit near-ideal autocorrelation, a sharp central peak with minimal sidelobes.
- Cross-correlation between different PRNs results in low-magnitude values spread across all chip delays.
- These values are small but **persistent**, even at zero Doppler.
- This persistence becomes significant when Doppler alignment slows their relative chip movement. (Spilker, 1996)

Relative Doppler and C/A Code Slippage

- The relative Doppler between two satellites determines how fast their PRN codes slip past each other.
- Code slip rate $= \dfrac{f_{\text{rel}}}{1540}$
- At 5 Hz Doppler, full slippage takes $\sim$308 seconds (5 minutes), causing long-lasting interference due to prolonged code overlap.
- At 1 kHz Doppler, slippage occurs in $\sim$1.5 seconds, creating short-lived but sharp spectral alignment (Spilker, 1996).
- Beach (2007) attributes lasting interference to slow chip movement at low relative Doppler.
- This explains how slowly evolving cross-correlation patterns can leave detectable signatures in frequency-domain data.

Data and Processing

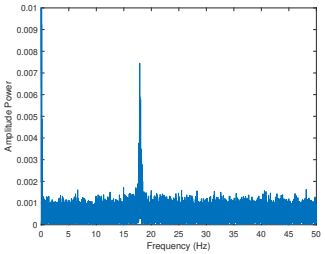
- Data obtained from CHAIN network's Septentrio PolarxS receivers at Churchill (58.75°N, 265.91°E).
- Observables sampled at 50 and 100 Hz, providing high-resolution phase and amplitude data.
- Detrending performed using a 0.1 Hz high-pass Butterworth filter to remove lower frequencies contributed from refraction.
- FFT applied to short time windows to extract dominant frequency variations.
- Doppler observable used to calculate relative Doppler between satellite pairs:

$$f_{\text{rel}} = f_i - f_j$$

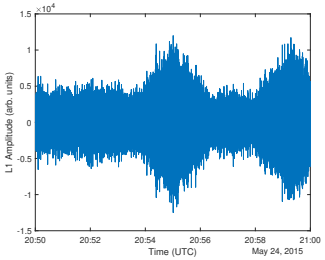
where f_i and f_j are Doppler values of satellites i and j.

Event Detection and Example

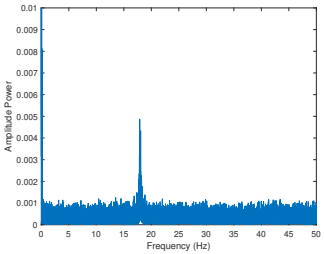
- Interference event detected using:
  - Dominant amplitude peaks exceeding mean $+ 2\sigma$ in two PRNs.
  - Time-domain fluctuations aligned across two PRNs within a ± 5 Hz Doppler-derived frequency range.
- These events sometimes repeat at the same sidereal time on consecutive days, similar to multipath effects.



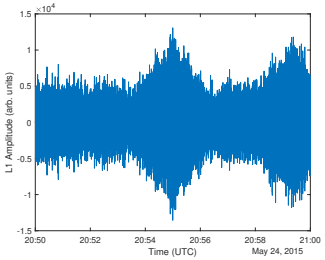
PRN 4 — Amplitude FFT



PRN 4 — Detrended Amplitude (24 May 2015)



PRN 11 — Amplitude FFT



PRN 11 — Detrended Amplitude (24 May 2015)

Distribution of Relative Doppler

- Histogram shows distribution of dominant frequencies across 218 interference events (2015).
- Bin size = 5 Hz; range spans 0–50 Hz.
- Events below 25 Hz are more frequent due to 50 Hz sampling constraint on some events. (Nyquist = 25 Hz).

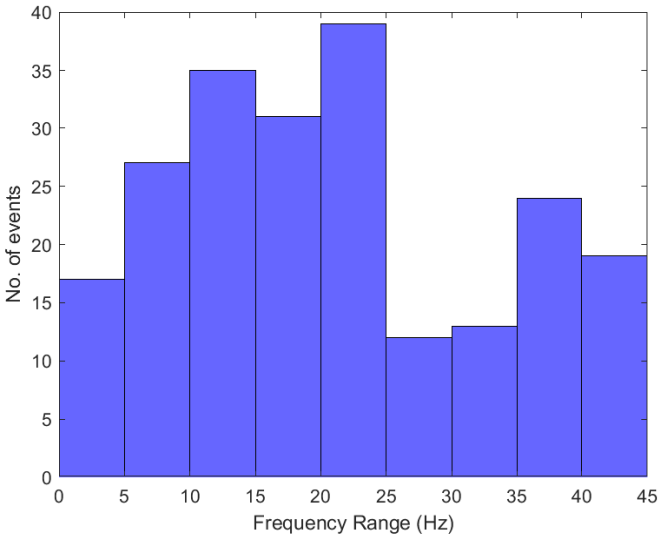
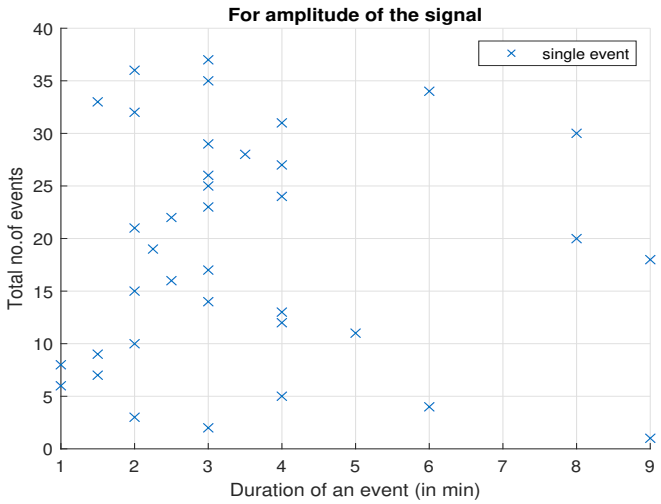


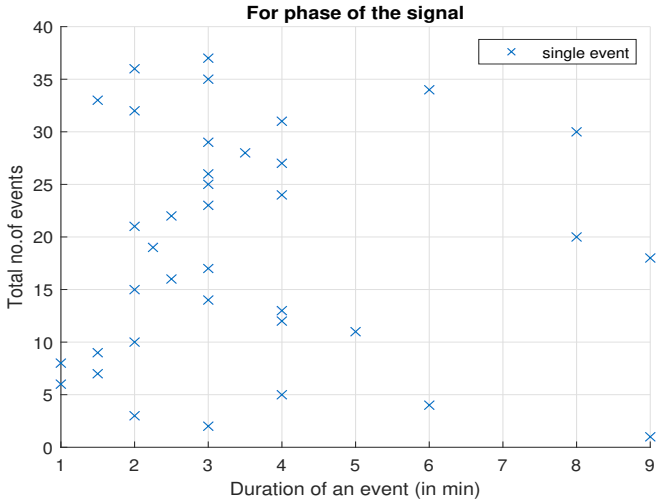
Fig: Histogram of relative Doppler values observed during interference events in 2015.

Duration of Interference Events

- Event durations identified using 2-minute separation buffer.
- Scatter plots show durations across 37 events for both amplitude and phase signals.
- Most events fall in the 1–3 minute range.
- Mean duration: 3.5 min; Mode and Median: 3 min.



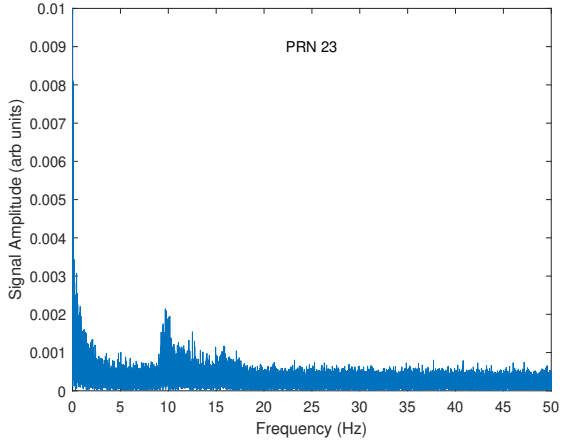
(a) Amplitude event durations



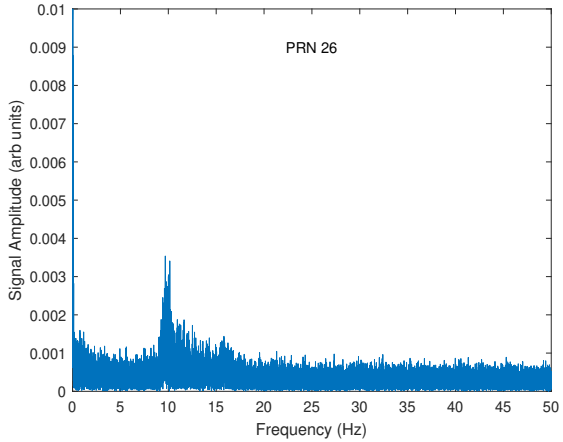
(b) Phase event durations

Aliased Interference Event Example

- Relative Doppler above Nyquist limit causes aliasing — high-frequency signals appear as lower frequencies.
- FFT peaks (10 Hz) correspond to true Doppler difference of 990 Hz.
- Aliased frequency: $f_{\text{observed}} = |f_{\text{true}} - Nf_s|$, where f_s is sampling rate and $N \in \mathbb{Z}$ keeps f_{observed} in-band.
- These aliased events were excluded from statistical analysis due to misrepresented frequency content.



(a) PRN 23 — 3 Oct 2015, 07:00 UTC



(b) PRN 26 — 3 Oct 2015, 07:00 UTC

Summary of Interference Characterization

- Amplitude and phase fluctuations resemble scintillation and multipath — making visual classification unreliable.
- Scintillation indices (S_4, σ_ϕ) show overlapping values across all three phenomena.
- Interference identified via matched peak frequencies and relative Doppler between PRNs.
- Event durations mostly ranged from 1–3 minutes — 1 min being comparable with typical scintillation bursts.
- Frequency analysis showed events across the Doppler spectrum, including aliased cases.
- Repeatable daily and sidereal patterns suggest geometry-driven interference, not noise.

These overlapping signal traits challenge traditional classification — motivating the need for modeling the phenomenon.

Power Expression for Interference (Beach 2007)

- Signal power observed while tracking PRN A, with PRN B acting as interferer:

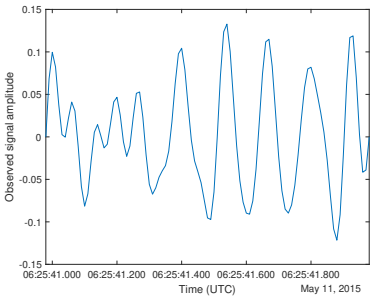
$$P_{An} = \frac{A_A^2}{2} + A_A A_B R_{BA}(\tau_A) \frac{\sin(\Omega_{BA} T_c/2)}{N \sin(\Omega_{BA} T_s/2)} \cos\left(\sum_{k=1}^{n-1} \Omega_{BA} T_c + \phi_B\right)$$

- $\Omega_{BA} = \Omega_B - \Omega_A$: relative Doppler between PRNs.
- $R_{BA}(\tau_A)$: PRN cross-correlation coefficient.
- A_A, A_B: signal amplitudes of satellites A and B.
- T_c, T_s: correlator and sampling intervals; N: number of samples.
- ϕ_B: Initial phase of PRN B at $t = 0$
- *Amplitude from power:* $\boxed{A = \sqrt{2P_{A_n}}}$

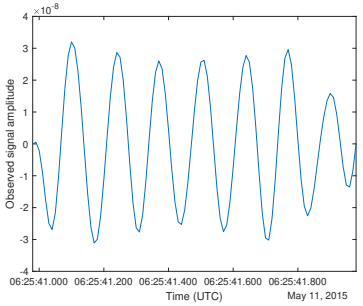
Model Validation: Observed vs Simulated Signal

Event Details

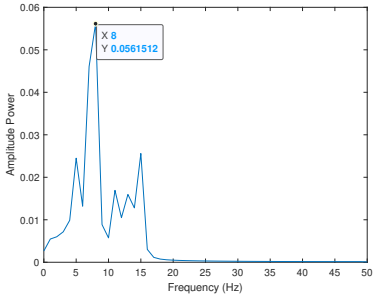
- Relative Doppler: 8 Hz
- Date: May 11, 2015 (06–07 UTC), PRN 5 and 20
- 1-second amplitude window extracted



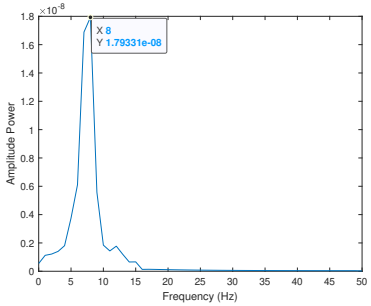
(a) Observed amplitude — time series



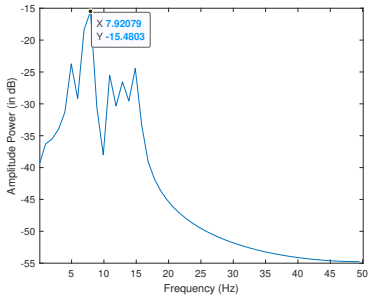
(b) Modeled amplitude — time series



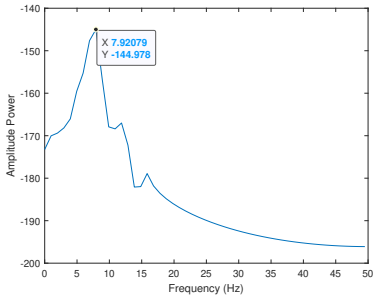
(c) Observed FFT



(d) Modeled FFT

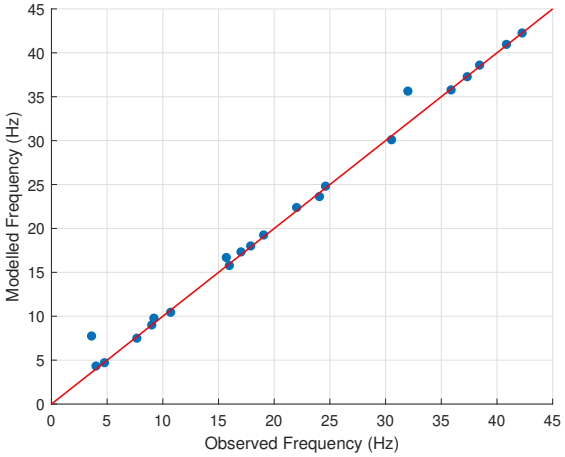


(e) Observed periodogram (dB)

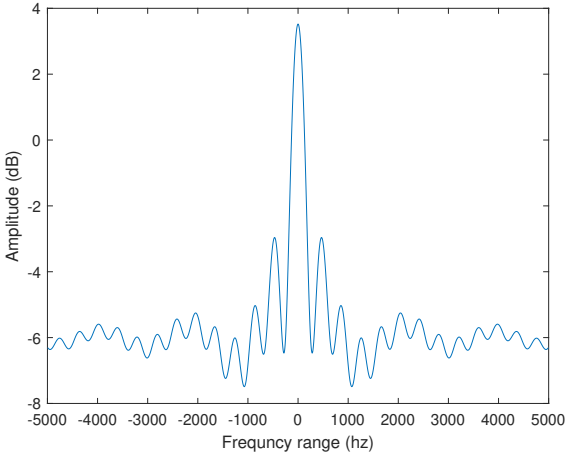


(f) Modeled periodogram (dB)

Further Analysis



(b) Distribution of events where observed peak frequencies match modeled ones



(a) Amplitude spectrum (dB) vs. Doppler frequency — Ratio of Sines behavior

Key Takeaways

- **Ratio of Sines:** Interference power is modulated by sinc-like behavior depending on $\Delta\omega_{BA}$.
- Interference peaks occur when Doppler separation is within ± 176 Hz (6 dB from peak).
- **Cross-Correlation:** Alignment of PRN codes at specific lags increases interference power.
- Doppler shifts cause time-varying alignment, making cross-correlation dynamic.
- **Combined Effect:** Fig. 5.3 shows strong match between modeled and observed peaks, validating both effects.

Together, these factors explain the amplitude and spectral patterns observed in real GPS signals.

Conclusion

- This study presents the first statistical and modeling-based analysis of inter-satellite interference caused by C/A code cross-correlation in GPS L1 signals.
- It identifies interference signatures in both time and frequency domains, revealing a predictable pattern tied to PRN code alignment and Doppler shifts.
- A simulation model derived from Beach (2007) was applied and compared with observed data from CHAIN receivers.
- The model distinguishes PRN-based interference from multipath and scintillation effects, where traditional metrics (e.g., S4, σ_ϕ) fail.
- Interference was observed across a wide range of PRNs, supporting its non-selective and constellation-wide nature.

Future Work

- Expand the dataset using multi-receiver and multi-constellation GNSS data to generalize findings.
- Implement higher sampling rates to minimize aliasing and improve frequency resolution in FFTs.
- Extend modeling to GLONASS, Galileo, and BeiDou to analyze cross-correlation effects with short-code formats.
- Use machine learning classifiers (e.g., SVM) to detect PRN interference based solely on FFT features.
- Integrate findings into GNSS receiver design to improve interference filtering and classification accuracy.
- Combine with GNSS simulator-based validation under controlled conditions.

Thank You!

Backup: Study Limitations

- Receiver sampling constraints introduced aliasing in some FFT outputs, particularly for high Doppler separation cases.
- The model focused on structural dynamics and did not incorporate real-time amplitude tracking, which could enhance future realism.
- Validation was limited to observed spectral patterns and did not involve hardware-in-the-loop testing or simulator replay under controlled conditions.

Backup: Cross-Correlation Properties of PRN Codes

Parameter	Code Period		
	511	**1023 (C/A)**	**2047**
Peak cross-correlation (any Doppler shift)	-18.6 dB	-21.6 dB	-24.6 dB
Peak cross-correlation (zero Doppler shift)	-23.8 dB	-23.8 dB	-29.8 dB
Probability of worst-case or near worst-case (zero Doppler)	0.5	0.25	0.5

Table: Cross-correlation properties of Gold codes of different lengths, adapted from Spilker (1996). C/A code used in GPS L1 has a 1023-chip length.

Backup: Colormap of Occurrence of Events

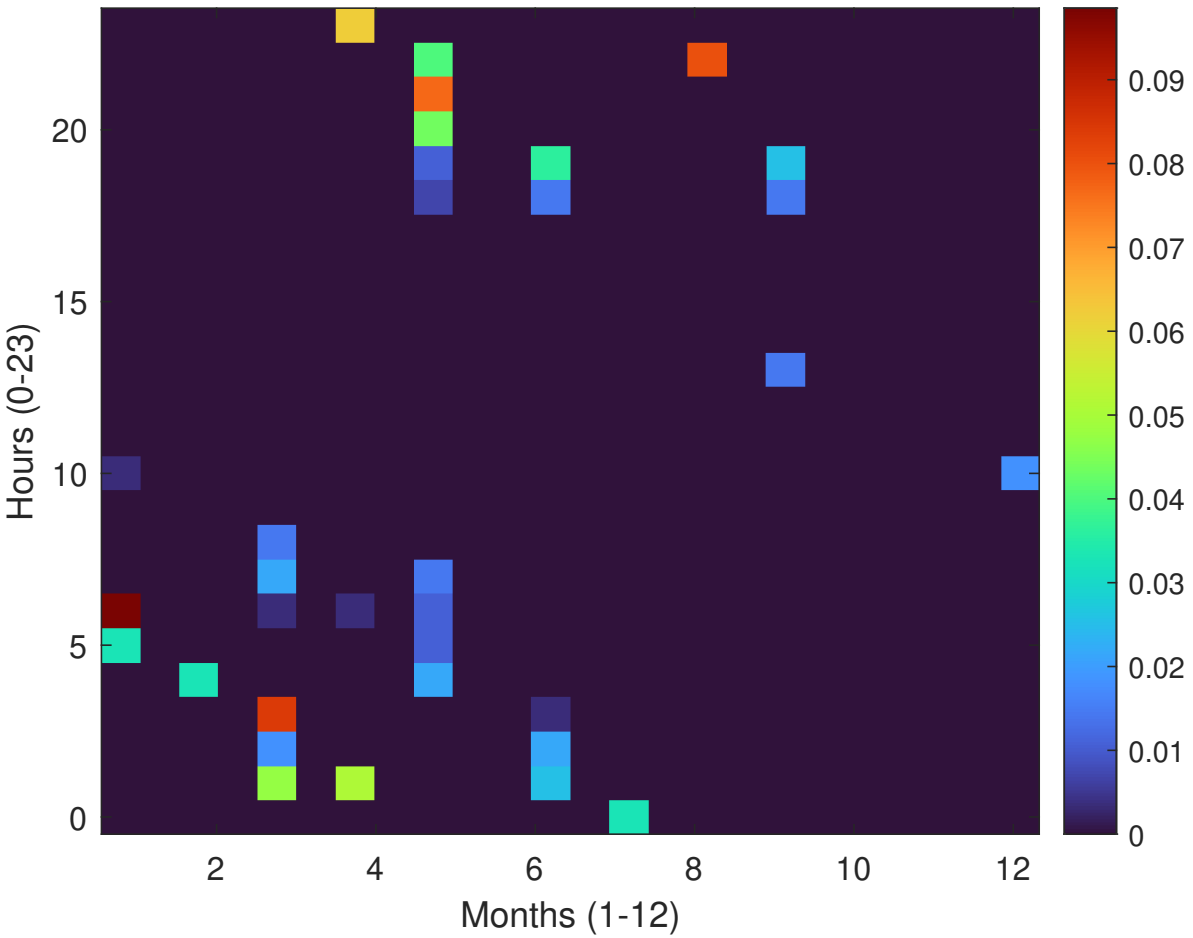


Figure: colormap showing the occurrence of PRN code interference events. Warmer regions correspond to higher event density.